Multi-tier Aggregation

1. What is Multi-tier Aggregation?

Multi-tier aggregation means using multiple layers of servers where each layer receives events from the previous layer, aggregates them further, and passes summarized data to the next layer.

Instead of one aggregation layer, the system uses several tiers.

2. How It Works

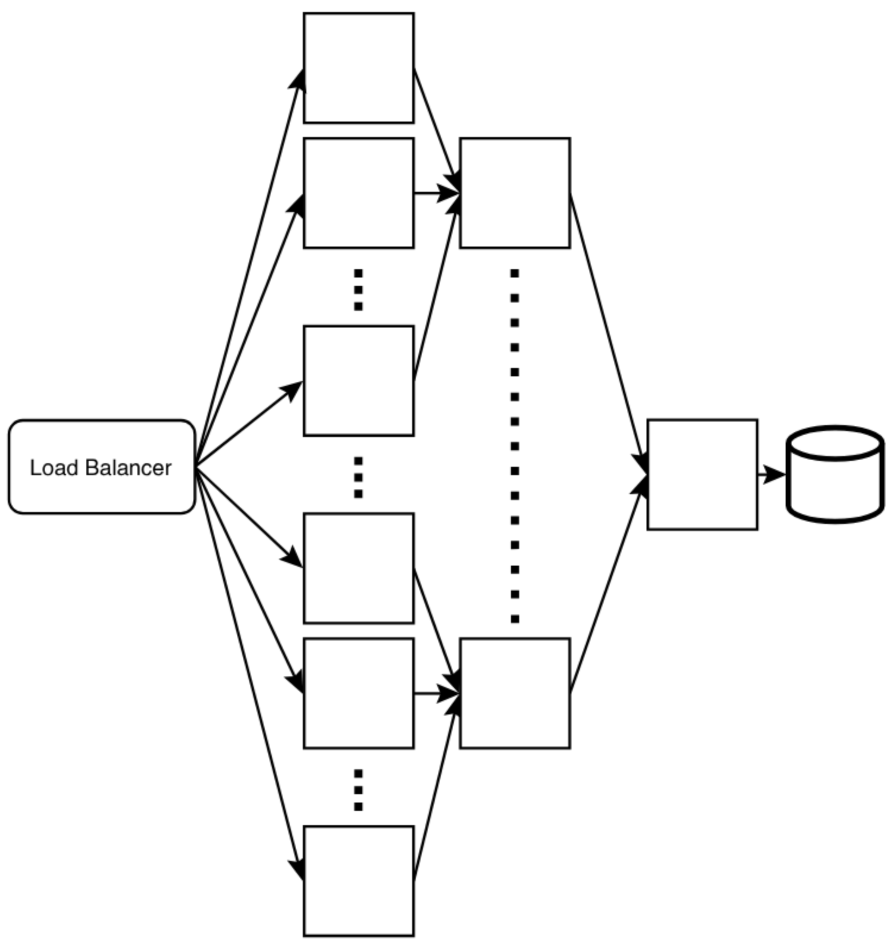

Figure 4.5 shows:

```None
Incoming Events

     ↓

Load Balancer

     ↓

Tier 1 Hosts

     ↓

Tier 2 Hosts

     ↓

Tier 3 Hosts

     ↓

Database
```

Each layer combines counts from the previous layer.

3. Example

Suppose raw events:

```None
A, B, B, C, C, D, B, C
```

Tier 1 hosts aggregate locally:

```None
Host 1:

A=1

B=2

C=1
```

```None
Host 2:

B=1

C=2

D=1
```

Tier 2 combines:

```None
A=1

B=3

C=3

D=1
```

Tier 3 final output:

```None
A=1

B=3

C=3

D=1
```

Then writes to database.

4. Why Use It

When incoming traffic is extremely high:

```None
Millions / billions of events per second
```

Single-tier may not be enough.

Multi-tier progressively reduces traffic at each stage.

5. Benefits

Without multi-tier:

```None
Every stage handles massive traffic
```

With multi-tier:

```
None
Tier 1 handles raw events

Tier 2 handles partial summaries

Tier 3 handles final summaries
```

Much smaller downstream load.

6. Host Reduction by Tier

Example:

```
None
Tier 1 = 1000 hosts

Tier 2 = 200 hosts

Tier 3 = 20 hosts

Final DB Writers = 5 hosts
```

Traffic shrinks every layer.

7. Real World Example

Ad click analytics:

```
None
1 Billion clicks/day
```

Tier 1 counts clicks per ad locally.

Tier 2 merges regional counts.

Tier 3 creates global totals.

DB stores only final aggregates.

8. Main Tradeoffs

```None
Higher latency

More stale reads

Operational complexity

Monitoring needed

Failure handling harder
```

9. Why Latency Increases

Each tier adds processing time:

```None
Tier 1 aggregate

→ send to Tier 2

→ aggregate

→ send Tier 3

→ aggregate

→ DB write
```

So final data may be delayed.

10. Fault Tolerance Needed

Need:

```None
Replication

Checkpointing

Retry queues

Monitoring

Alerting

Logging
```

11. Best Use Cases

- Search analytics
- Clickstream processing
- Sensor networks
- Ad impressions
- Social media reactions
- IoT telemetry

12. When Not to Use

Avoid for:

```None
Payments

Banking ledgers

Low-latency exact transactions

Audit systems
```

13. Interview Talking Point

If traffic is massive:

```None
Use multi-tier aggregation

Reduce traffic layer by layer

Few final DB writes

Better scalability
```

Strong system design answer.

14. One-Line Summary

Multi-tier aggregation uses multiple layers of aggregators to progressively combine high-volume events before writing final summaries to the database, improving scalability at the cost of latency and complexity.